

CHSH Bell Inequality Test

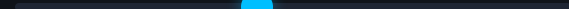
Monte-Carlo of the singlet state $|\Psi^-\rangle$ at four measurement angles (a, a', b, b'): the running estimate $\hat{S}$ approaches the Tsirelson bound $2\sqrt{2} \approx 2.828$ at the canonical Bell angles (0°, 90°, 45°, -45°), violating the local-realist limit $|S| \leq 2$ — quantum entanglement made statistically visible.

Pause sampling



SINGLET $|\Psi^-\rangle$ + MEASUREMENT ANGLES

Alice angle a -25°



Alice angle a' -10°

Bob angle b 55°

Bob angle b' 180°

Trials / frame 246

Bell preset ($S = 2\sqrt{2}$)

Classical ($S = 2$)

Reset counts

Two-qubit singlet $|\Psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$: $\langle \sigma_a \cdot \sigma_b \rangle = -\cos(a-b)$. The CHSH combination $S = E(a,b) - E(a,b') + E(a',b) + E(a',b')$ saturates the local-realist bound at $|S| \leq 2$ and the quantum (Tsirelson) maximum at $|S| = 2\sqrt{2} \approx 2.828$, reached at the canonical Bell angles $(0^\circ, 90^\circ, 45^\circ, -45^\circ)$.

MEASURED VALUES	
$\hat{S}$ (Monte-Carlo)	-0.521
S_{theory}	-0.518
Trials	64206
Violates $ S \leq 2$?	no

About this model

This lab runs a Monte-Carlo CHSH Bell test on the singlet state $|\Psi^-\rangle$. Four measurement settings (a, a', b, b') define correlators that combine into the CHSH parameter $\hat{S}$. At the canonical Bell angles (0°, 90°, 45°, -45°) the running estimate approaches the Tsirelson bound $2\sqrt{2} \approx 2.828$, violating the local-realist limit $|S| \leq 2$. Each trial samples projective outcomes ± 1 according to quantum probabilities for the singlet. The simulator assumes ideal projective measurements, perfect entanglement, and no detection loopholes or communication between parties. You choose angles and accumulate shots to watch $\hat{S}$ converge — entanglement made statistically visible against the classical bound.

Who it's for: Quantum information and foundations courses introducing Bell inequalities, entanglement, and local realism.

KEY TERMS

- CHSH inequality
- Bell test
- Singlet state
- Tsirelson bound
- Entanglement
- Local realism

How it works

CHSH Bell test simulator: a Monte-Carlo of two entangled spins in the singlet state shows the running estimate $\hat{S}$ approaching the quantum limit $2\sqrt{2} \approx 2.828$ at the canonical angles (0°, 90°, 45°, -45°), well above the local-realist bound $|S| \leq 2$.

Frequently asked questions

What does violating $|S| \leq 2$ prove?

Any local-hidden-variable model that assigns predetermined outcomes independent of the distant setting cannot exceed $|S| = 2$. Quantum mechanics predicts up to $2\sqrt{2}$ for the singlet at optimal angles. Observing $\hat{S} > 2$ (with enough statistics) rules out that classical picture for the prepared state — it does not by itself close experimental loopholes.

Why is $2\sqrt{2}$ the maximum, not 4?

Algebraically each correlator is at most 1, so $|S| \leq 4$ looks possible, but quantum measurements on two qubits cannot saturate that. Tsirelson’s bound $2\sqrt{2}$ is the tight quantum maximum for CHSH; reaching 4 would require stronger-than-quantum correlations.

Do I need exactly the angles $0^\circ, 90^\circ, 45^\circ, -45^\circ$?

Those angles maximize the singlet violation. Other choices can still violate $|S| \leq 2$, but with a smaller margin; some angles give S compatible with local realism even for an entangled state. The lab lets you mis-set angles to see the violation shrink or disappear.

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